Combinatorics 2024 Final Solutions

1. Use the binomial theorem to prove that

$$2^n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} 3^{n-k}.$$

Solution

We have that

$$2^n = (3-1)^n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} 3^{n-k}.$$

2. Consider the poset $2^{[n]}$ of subsets of $[n]$ ordered by inclusion.

   (a) How many longest chains are there in the poset?

   (b) How many longest chains are there containing a given set S of size k?

   (c) How many chains of length k in the poset contain contain both the empty set and the set $[n]$? (You haven't seen this one before, but you can figure it out.)

Solution

i) We get a longest chain by starting at the empty set and adding one element at at time. We can do this in $\boxed{n!}$ ways. ii) For a longest chain containing S we have to add the elements of S first. We can do this in $\boxed{k!(n-k)!}$ ways. iii) For a chain of length k, we have to add the elements in $k-1$ subsets. So we are counting the number of ways to distribute n distinguishable elements into $k-1$ non-empty distinguishable bins. This is would be $(k-1)^n$ ways if it were not for the 'non-empty', but we have to deal with this, so we use the PIE. Let S be the distributions of n distinguishable elements into $k-1$ distinguishable bins, and for $i = 1, \ldots, k-1$ let A_i be the number of such distributions in which bin i is empty. By the PIE, the number of distributions with no empty bins

$$|S - (\bigcup A_i)| = \sum_{i=0}^{k-1} (-1)^i \sum_{I \subset \binom{[k-1]}{i}} a_I = \boxed{\sum_{i=0}^{k-1} (-1)^i \binom{k-1}{i} (k-1-i)^n}.$$

3. How many permuations of [10] fix at least one odd number?

Solution

Where A_i is the set of permutations that fixes i we use the dual form of PIE to count $|A_1 \cup A_3 \cup \cdots \cup A_5|$. For $I \subset \{1,3,5,7,9\}$ where we have that a_I counts the permutations that fixes the points in I, so is $(10-|I|)!$. There are $\binom{5}{i}$ subsets I of size i, so the total number of such permutations is

$$\sum_{i=1}^{5}(-1)^{i+1} \sum_{I \subset \binom{\{1,3,5,7,9\}}{i}} a_I = \boxed{\sum_{i=1}^{5}(-1)^{i-1}\binom{5}{i}(10-i)!} =$$

$$\binom{5}{1}\cdot 9! - \binom{5}{2}\cdot 8! + \binom{5}{3}\cdot 7! - \binom{5}{4}\cdot 6! + \binom{5}{5}5!.$$

A lot of people tried to just tried 10! minus the permutions that fix no odd numbers, and counted this as $D_5 \cdot 5!$. But $D_5 \cdot 5!$ counts the permutations that fixes no odd numbers, but also keeps odd number in odd spots and even number in even spots.

4. How many permutations of $[n]$ have at least two fixed points? (You may use D_n to denote the number of derangements of $[n]$.)

Solution

The number that fix of permutations of $[n]$ that fix not points is D_n and the number that fix exactly one point is $n \cdot D_{n-1}$ as we have n ways to choose the fixed point, and then D_{n-1} ways to arrange the other points with no fixed points. So there are $\boxed{n! - D_n - nD_{n-1}}$ such permutations.

5. Where $f_0, f_1, f_2, \ldots$ are the Fibonacci numbers, show that the partial sum $s_n = \sum_{i=0}^{n} f_i$ is equal to $f_{n+2} - 1$.

Solution

We use induction. As $s_0 = 0 = 1 - 1 = f_2 - 1$, the identity holds for $n = 0$ which gives us the base case. Assuming $n > 0$ and that it holds for all values less than n, we see

$$\begin{aligned} s_n &= s_{n-1} + f_n \\ &= f_{n+1} - 1 + f_n \\ &= f_{n+2} - 1 \end{aligned}$$

6. Solve the recurrence relation $h_n = 3h_{n-1} - 2h_{n-2}$ for $n \geq 2$ with initial values $h_0 = -1$ and $h_1 = 0$.

Solution

Guessing a solution of the form $h_n = q^n$ we get from the recurrence that $q^n = 3q^{n-1} - 2q^{n-2}$. This holds if and only if

$$0 = q^{n-2}(q^2 - 3q + 2) = q^{n-1}(q-1)(q-2),$$

which means that $q = 0, 1$ or 2. Any function $h_n = a1^n + b2^n$ satisfies the recurrence, and the initial conditions give $-1 = h_0 = a+b$ and $0 = h_1 = a+2b$ which we solve to get $a = -2$ and $b = 1$. Thus $h_n = 2^n - 2$.

7. Using an exponential generating function, find the number h_n of ways of colouring the squares of a $1 \times n$ chessboard with 4 colours red, yellow, blue and white, so that there are an even number of red and blue tiles.

Solution

The exponential generating function for $h_0, h_1, h_2, \dots$ is

$$\begin{aligned} g^{(e)}x &= (1 + x + \frac{x^2}{2!} + \dots)^2 (1 + \frac{x^2}{2!} + \dots)^2 \\ &= e^{2x}\left(\frac{e^x + e^{-x}}{2}\right)^2 = \frac{1}{4}(e^{4x} + 2e^{2x} + 1) \\ &= \frac{1}{4}\left((1 + 4x + 4^2\frac{x^2}{2!} + 4^3\frac{x^3}{3!}\dots) + 2(1 + 2x + 2^2\frac{x^2}{2!} + 2^3\frac{x^3}{3!}) + 1\right) \\ &= 1 + x + (2+4)\frac{x^2}{2!} + (2^2 + 4^2)\frac{x^3}{3!} + \dots \end{aligned}$$

From this we read off that $h_n = 2^{n-1} + 4^{n-1}$ for $n \geq 1$. And $h_0 = 1$.

8. Let $h_n = f(n)$ where $f(n)$ is a polynomial of degree 3, and the 0th row of the difference table for $(h_0, h_1, \dots)$ is $(1, -1, 3, 10, \dots)$. Find a formula for $\sum_{i=1}^{n} h_i$.

Solution

The difference table is

1		-1		3		10		...	
	-2		4		7		...		
		6		3		...			
			-3		...				
				0		0	...		

The 4th row is a row of zeros because $f(n)$ has degree 3. Our formula for h_n in terms of the entries of the 0th diagonal gives:

$$h_n = \binom{n}{0} - 2\binom{n}{1} + 6\binom{n}{2} - 3\binom{n}{3}.$$

To find the sum $\sum_{i=1}^{n} h_i$ we use the identity $\binom{n+1}{k+1} = \binom{n}{k} + \binom{n-1}{k} + \binom{n-2}{k} + \cdots + \binom{0}{k}$. We get

$$\sum_{i=0}^{n} h_i = \binom{n+1}{1} - 2\binom{n+1}{2} + 6\binom{n+1}{3} - 3\binom{n+1}{4},$$

and so (most people missed this tricky little accident of mine)

$$\boxed{\sum_{i=1}^{n} h_i = \binom{n+1}{1} - 2\binom{n+1}{2} + 6\binom{n+1}{3} - 3\binom{n+1}{4} - 1.}$$

9. For each integer $n \geq 5$ find a self-conjugate partition of n with at least 3 parts.

Solution

If n is odd, let $\ell = (n-1)/2$; the partition $\ell + 1 \ldots 1$ with ℓ ones is self-conjugate. If n is even let $\ell = n/2$, the partition $\ell + 2 + 1 + \cdots + 1$ with $\ell - 2$ ones is self-conjugate.